Reliability Evaluation of MQTT-SN over Wireless Sensor Networks

Using RSSI-Driven Adaptive Sampling on ESP32 IoT Nodes

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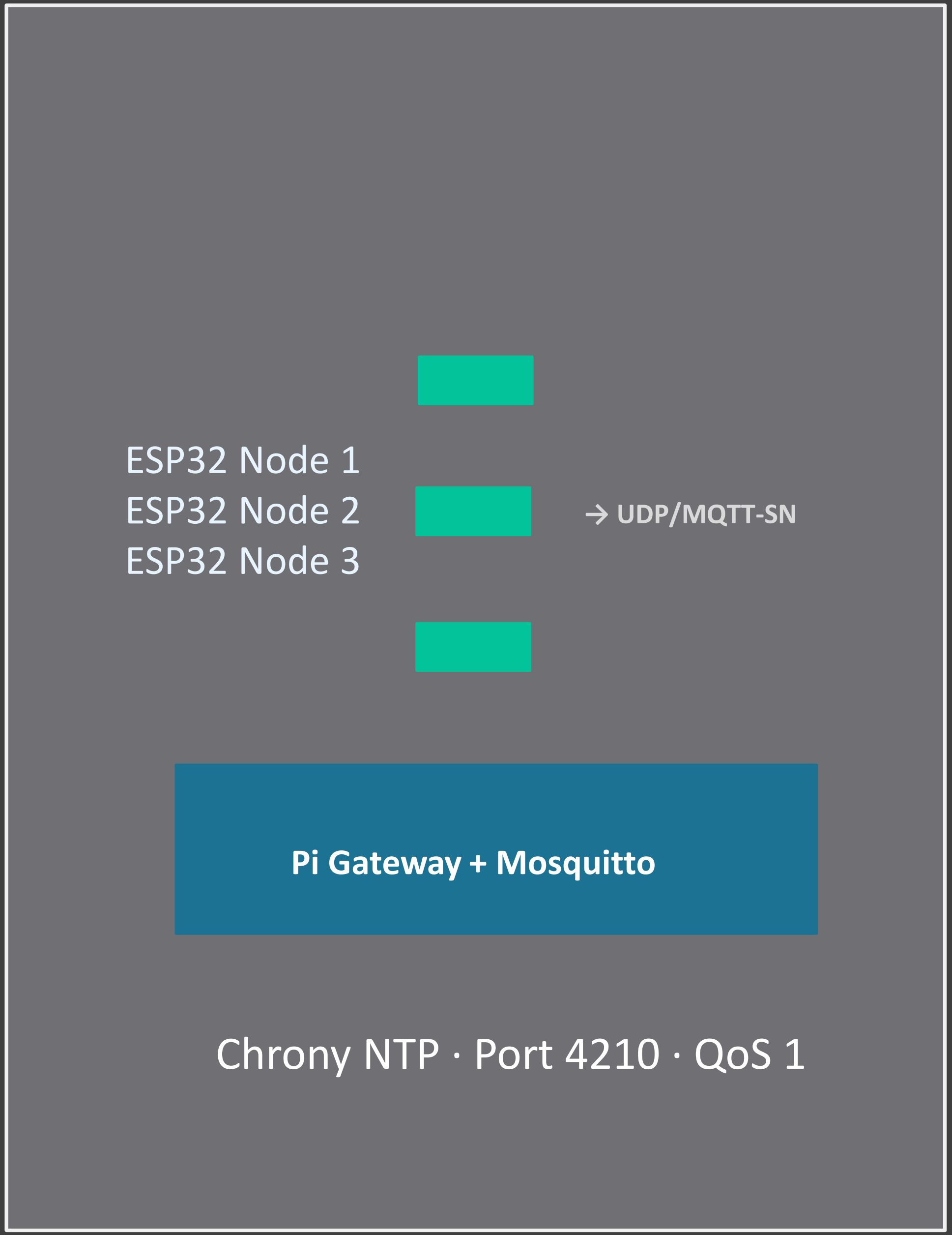
Raspberry Pi 4B Gateway – 3xESP32-WROOM-DA Nodes

Problem & Motivation

- Standard MQTT uses TCP → unpredictable retransmission latency
- MQTT-SN uses UDP → lightweight, suitable for constrained IoT nodes
- Fixes publish rate + weak Wi-Fi signal = packet collisions & drops
- **Key question: Can RSSI driven rate adaptation improve reliability?**



Component	Detail
Gateway	Raspberry Pi 4B – Mosquitto 2.0.21, chrony NTP
Sensor Nodes	3x ESP32-WROOM-DA, 240MHz
Protocol	MQTT-SN over UDP → MQTT (QoS 1)
Transport	IEEE 802.11n, 2.4 GHz, port 4210
Baseline Rate	10 Hz (100ms interval)
Adaptive Range	0.5-20Hz (RSSI-driven)
Node ID	MAC address lookup table

Testbed & Experiment Setup

Adaptive Sampling Algorithm

RSSI-Driven Interval Selection

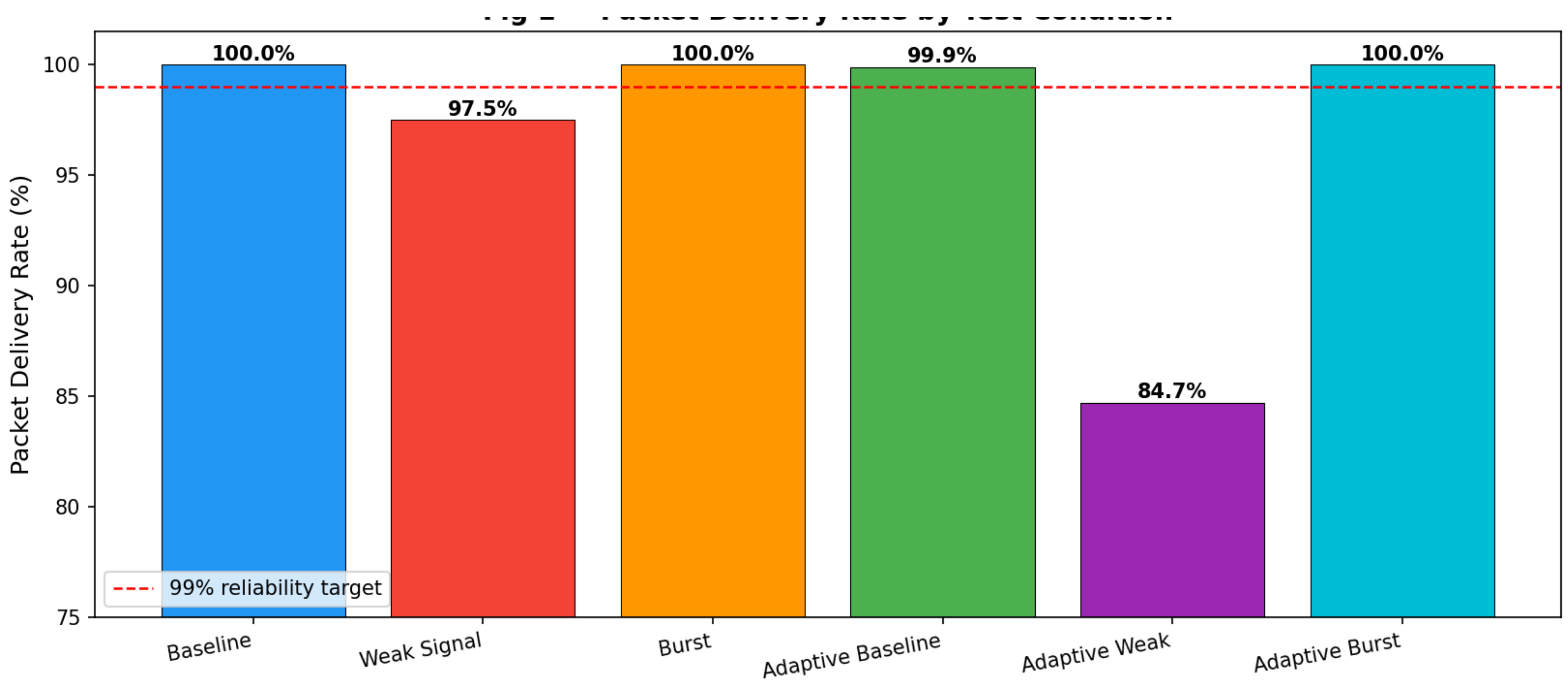
- Measure RSSI every transmission cycle
- Look up interval from 4-tier table
- Transmit, then wait the selected interval
- **Result: nodes bac off automatically when channel degrades**

- Excellent (≥-60)
- Good (≥-70)
- Weak (≥-80)
- Poor (<-80)

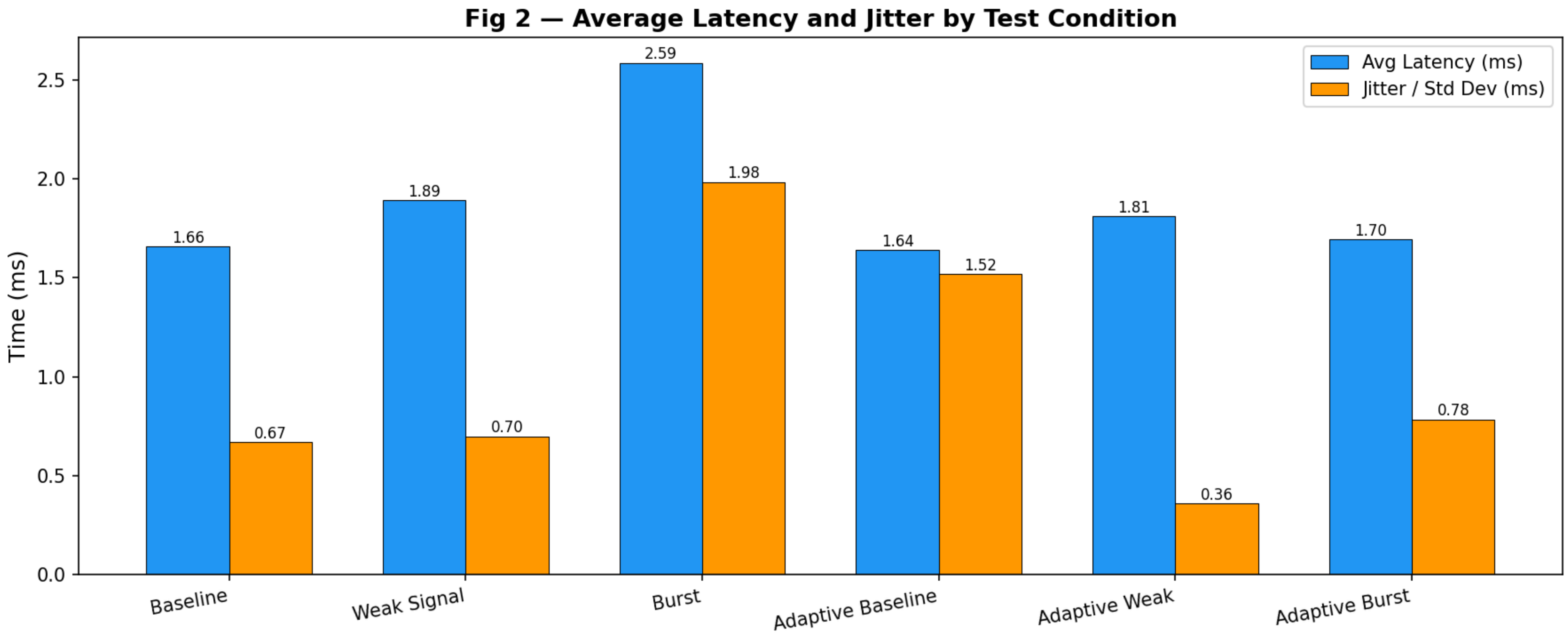
RSSI Range	Interval	Rate
≥ -60dBm	50ms	20Hz
≥ -70dBm	100ms	10Hz
≥ -80dBm	500ms	2Hz
< -80dBm	2000ms	0.5Hz

Experimental Conditions

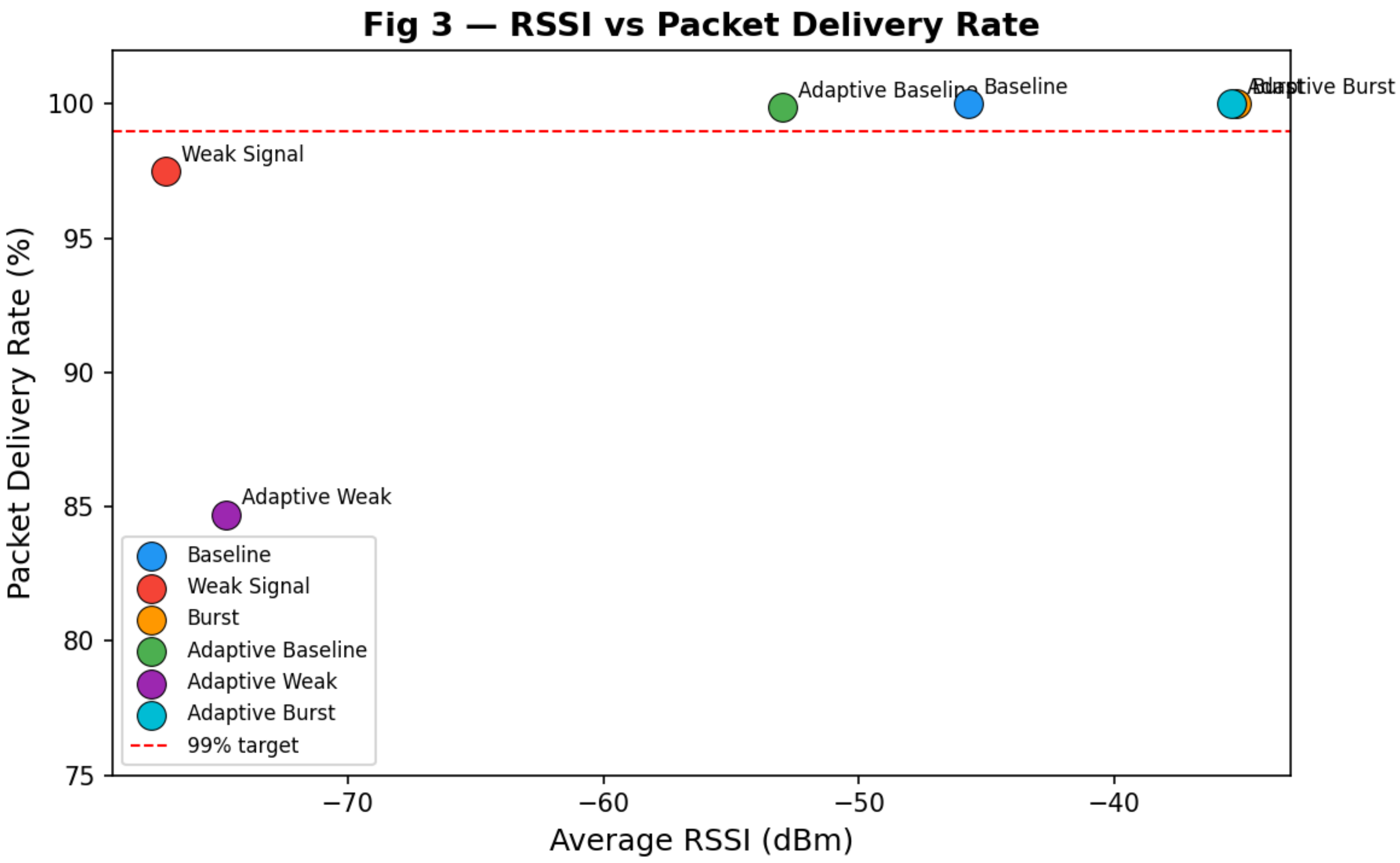
#	Condition	Sketch	Hotspot	RSSI
1	Baseline	Steady 10Hz	Co-located	-45.7dBm
2	Weak	Steady 10Hz	Adjacent room	-77.1dBm
3	Burst	Steady 10Hz	Co-located	-35.2dBm
4	Adaptive Baseline	Adaptive	Co-located	-53.0dBm
5	Adaptive Weak	Adaptive	Adjacent room	-74.7 dBm
6	Adaptive Burst	Adaptive	Co-located	-35.4dBm



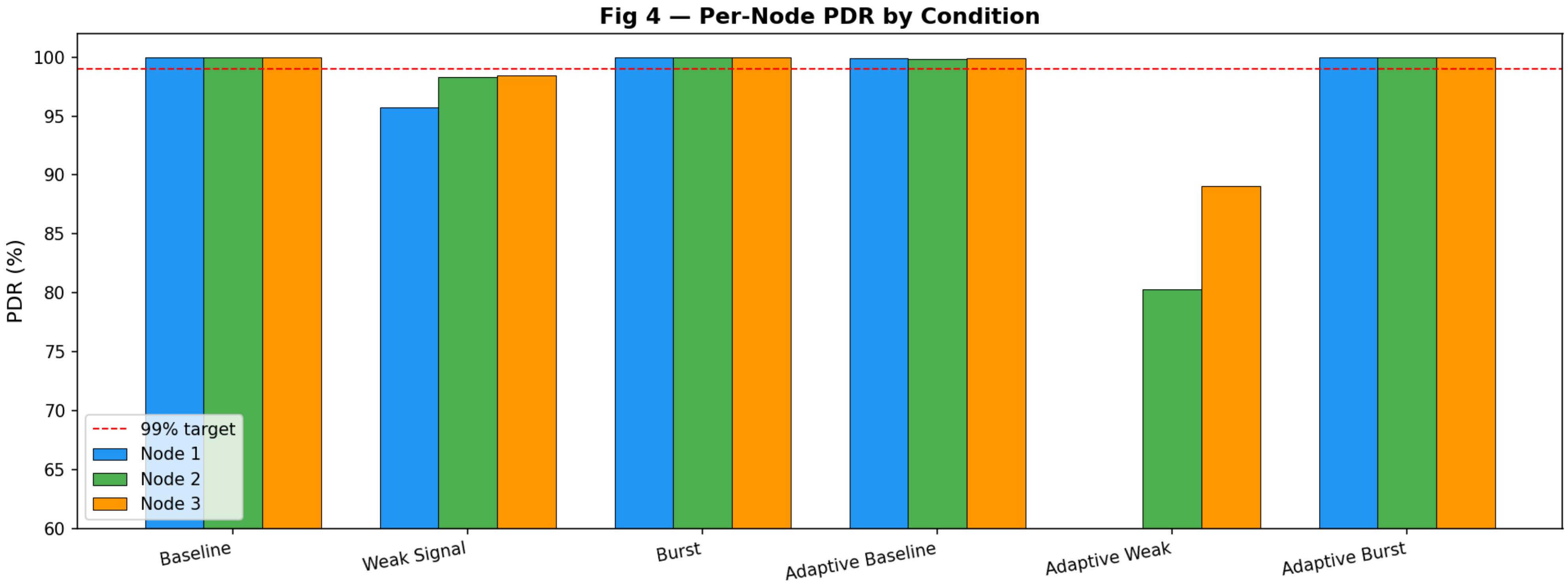
Results – Packet Delivery Rate



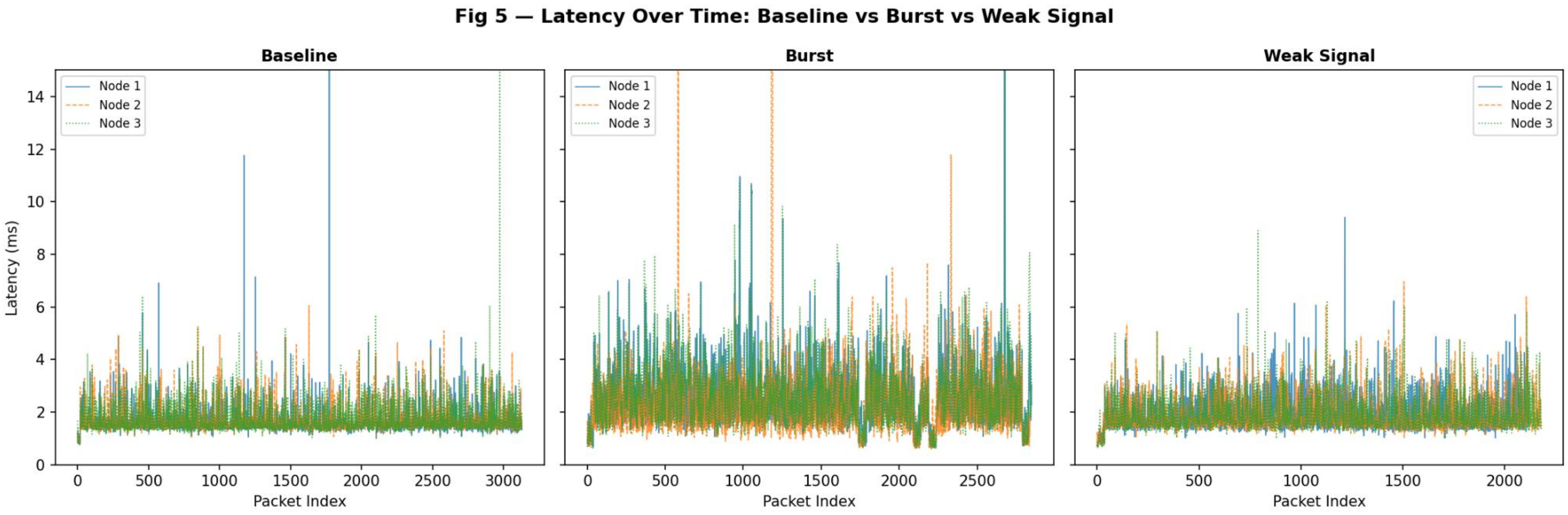
Results – Latency & Jitter



Results – Signal Quality vs Reliability



Results – Per-Node Breakdown



Result – Latency Over Time

Summary of Results

Condition	PDR (%)	Avg Latency (ms)	Jitter (ms)	Avg RSSI (dBm)	Drops
Baseline	100.0	1.66	0.67	-45.7	0
Weak	97.5	1.89	0.70	-77.1	167
Burst	99.9	1.64	1.52	--53.0	19
Adaptive Baseline	84.7	1.81	0.36	-74.7	152
Adaptive Weak	100.0	2.59	1.98	-35.2	0
Adaptive Burst	100.0	1.70	0.78	-35.4	0

Key Findings

MQTT-SN achieves 100% PDR and sub-2ms latency under strong signal conditions

Weak signal (RSSI < -70dBm) reduces PDR to 97.5% under fixed rate transmission

Burst traffic does not degrade reliability when channel quality is high (100% PDR)

Adaptive sampling reduces jitter to 0.36ms under weak signal – lowest of all test

RSSI and PDR show clear positive correlation across all test conditions

Limitations & Future Reference

Limitations

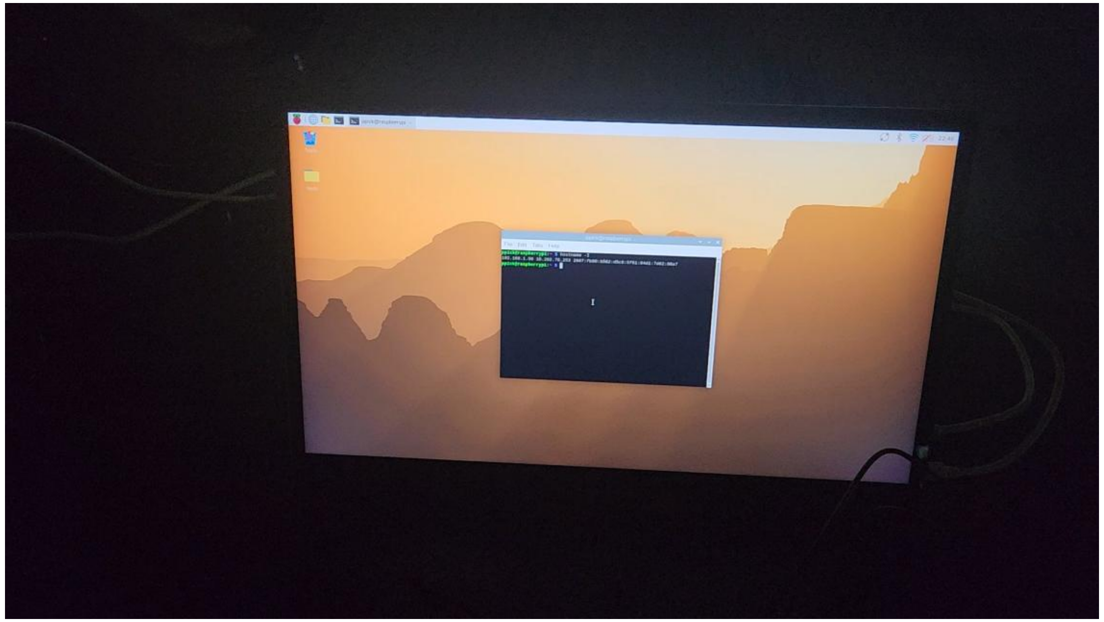
- NTP sync accuracy (1-5ms) comparable to measured latency
- Adaptive algorithm has no hysteresis → rapid switching near thresholds
- Adaptive-weak run missing Node 1 (network dropout)
- Single AP - no multi-hop or mesh evaluation

Future Work

- Hardware PTP for sub-millisecond latency measurement
- RSSI exponential moving average for smoother adaptation
- Extend to IEEE 802.15.4 (Zigbee) nodes for protocol comparison
- Test under controlled interference (microwave, BT)

Conclusion and Demo example

- MQTT-SN is a viable, low latency protocol for Wi-Fi IoT sensor networks
- Sub-2ms latency and 100% PDR achievable under strong signal
- **Adaptive sampling reduces jitter under degraded RF conditions**
- RSSI is a reliable predictor of delivery reliability



Demo of an adaptive-burst MQTT-SN for broker and with chrony for network time protocol for logging that saves .csv to the Pi